SPAM EMAIL CLASSIFICATION USING MACHINE LEARNING

Project Report

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Abstract

Spam emails and messages are one of the most common cybersecurity and communication problems today. These unwanted messages often contain advertisements, phishing links, scams, or malicious content. Manually filtering spam messages is inefficient and time-consuming.

This project presents a Machine Learning based spam email classification system capable of automatically identifying spam and non-spam messages using Natural Language Processing (NLP) techniques. The system uses TF-IDF vectorization to convert textual data into numerical format and applies multiple Machine Learning algorithms including Naive Bayes, Logistic Regression, and Support Vector Machine (SVM) for classification.

The models were trained and tested using the SMS Spam Collection Dataset. Comparative analysis showed that the Support Vector Machine achieved the highest accuracy. The project demonstrates the effectiveness of NLP and Machine Learning in automated text classification systems.

1. Introduction

Electronic communication platforms are widely used for personal and professional communication. However, the rapid increase in spam messages has created security and productivity challenges. Spam messages often contain fraudulent advertisements, phishing attempts, malware links, or misleading information.

Spam detection systems are important because they help users:

- avoid malicious content
- reduce unwanted messages
- improve communication security
- save time and storage resources

Machine Learning provides an efficient solution for spam detection by automatically learning patterns from previously labeled messages. This project focuses on building an intelligent spam classification system using Machine Learning and Natural Language Processing techniques.

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2. Objectives

The main objectives of this project are:

- To build a Machine Learning model for spam email classification
- To preprocess and clean textual data using NLP techniques
- To convert text into numerical features using TF-IDF vectorization
- To compare multiple Machine Learning algorithms
- To evaluate model performance using accuracy and confusion matrix
- To perform real-time spam prediction on custom messages

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3. Technologies Used

- Python
- Google Colab
- Pandas
- NumPy
- Scikit-learn
- Natural Language Processing (NLP)
- TF-IDF Vectorization
- Naive Bayes
- Logistic Regression
- Support Vector Machine (SVM)

- Matplotlib
- Seaborn
- WordCloud
- Pickle

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4. Dataset Description

The project uses the SMS Spam Collection Dataset obtained from the UCI Machine Learning Repository.

Dataset Link:
SMS Spam Collection Dataset

Dataset Download Link:
Download Dataset ZIP File

Dataset Features:

Feature	Description
label	spam or ham
message	actual text message

Dataset Information:

- Total records: 5572
- Spam messages: 747
- Non-spam messages: 4825

The dataset contains real SMS messages labeled as spam or non-spam.

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5. Methodology

The project follows several stages for spam classification.

5.1 Data Collection

The SMS Spam Collection Dataset was imported into Google Colab using Pandas.

5.2 Data Preprocessing

Text preprocessing was performed to improve model performance. The following operations were applied:

- Conversion to lowercase
- Removal of punctuation
- Removal of numbers
- Removal of URLs

- Removal of extra spaces

5.3 Exploratory Data Analysis (EDA)

EDA techniques were used to understand the dataset distribution and text characteristics.

The following visualizations were generated:

- Spam vs Ham count graph
- Spam WordCloud
- Ham WordCloud

5.4 Feature Extraction

Machine Learning models cannot process raw text directly. Therefore, TF-IDF Vectorization was used to convert text into numerical vectors.

TF-IDF measures the importance of words in the dataset while reducing the influence of commonly occurring words.

5.5 Model Training

Three Machine Learning algorithms were used:

Naive Bayes

A probabilistic classification algorithm suitable for text classification problems.

Logistic Regression

A supervised learning algorithm used for binary classification tasks.

Support Vector Machine (SVM)

A high-performance classification algorithm effective for high-dimensional textual data.

5.6 Model Evaluation

The models were evaluated using:

- Accuracy Score
- Classification Report
- Confusion Matrix

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6. System Workflow

Dataset
↓
Data Cleaning
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Text Preprocessing
↓
TF-IDF Vectorization
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Train-Test Split

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Machine Learning Models
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Prediction
↓
Performance Evaluation

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7. Results and Analysis

The project successfully classified spam and non-spam messages with high accuracy.

Approximate model accuracies:

Model	Accuracy
Naive Bayes	97%
Logistic Regression	96%
SVM	98%

Among all tested algorithms, SVM achieved the best performance for spam classification.

The confusion matrix and classification report demonstrated strong precision and recall values, indicating reliable prediction performance.

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11. Conclusion

This project successfully developed a Machine Learning based spam email classification system using Natural Language Processing techniques. The system effectively classified spam and non-spam messages using TF-IDF vectorization and multiple Machine Learning algorithms.

Comparative analysis showed that the Support Vector Machine provided the highest classification accuracy. The project demonstrates the practical application of Machine Learning in cybersecurity and automated communication systems.

The developed system is efficient, scalable, and suitable for real-world spam filtering applications.

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